son.pu.py - C:/Users/pings/Desktop/son.pu.py (3.14.2)

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#Sampling
#Program 2: Sampling Distribution and Standard Error of the Mean
#Aim: To build the sampling distribution of the sample mean and compute the Standard Error (SE) of the mean for different sample sizes.
#Scenario: A courier company in Chennai records the delivery time (minutes) of 1,00,000 parcels(population mean about 40 minutes, SD about 8 minutes).
#The manager takes 1000 random samples each of size n = 10, 30 and 100, finds the mean of every sample, and studies how these sample means are spread.

import numpy as np
import matplotlib.pyplot as plt
np.random.seed(1)
# Step 1: Population of delivery times (minutes)
population = np.random.normal(loc=40, scale=8, size=100000)
mu, sigma = population.mean(), population.std()
print(f"Population mean = {mu:.2f} min, Population SD = {sigma:.2f} min\n")
# Step 2: Take 1000 samples for each sample size and store their means
sample_sizes = [10, 30, 100]
num_samples = 1000
fig, ax = plt.subplots(2, 2, figsize=(12, 8))
ax = ax.ravel()
ax[0].hist(population, bins=50, color='lightgray', edgecolor='black')
ax[0].axvline(mu, color='red', linestyle='--', label=f'Population mean = {mu:.1f}')
ax[0].set_title('Population: Delivery Times of 1,00,000 Parcels')
ax[0].set_xlabel('Minutes');
ax[0].set_ylabel('Frequency');
ax[0].legend()
print(f"{'n':>5} {'Mean of sample means':>22} {'SE = σ/√n':>12} {'SD of sample means':>20}")
for i, n in enumerate(sample_sizes):
    sample_means = [np.random.choice(population, size=n, replace=False).mean() for _ in range(num_samples)]
    se_formula = sigma / np.sqrt(n)
    se_observed = np.std(sample_means)
    print(f"{n:>5} {np.mean(sample_means):>22.2f} {se_formula:>12.2f} {se_observed:>20.2f}")
    ax[i+1].hist(sample_means, bins=30, color='lightgreen', edgecolor='black')
    ax[i+1].axvline(mu, color='red', linestyle='--', label='Population mean')
    ax[i+1].set_xlim(30, 50) # same scale so the spread can be compared
    ax[i+1].set_title(f'Sampling Distribution of Mean (n = {n}), SE = {se_formula:.2f}')
    ax[i+1].set_xlabel('Sample mean (minutes)');
    ax[i+1].set_ylabel('Frequency');
ax[i+1].legend()
plt.tight_layout()
plt.show()
# Step 3: SE estimated from ONE sample of 30 parcels (sigma not known)
one_sample = np.random.choice(population, size=30, replace=False)
se_est = one_sample.std(ddof=1) / np.sqrt(30)
print(f"\nOne sample of 30 parcels: mean = {one_sample.mean():.2f}, estimated SE = s/√n = {se_est:.2f}")
```

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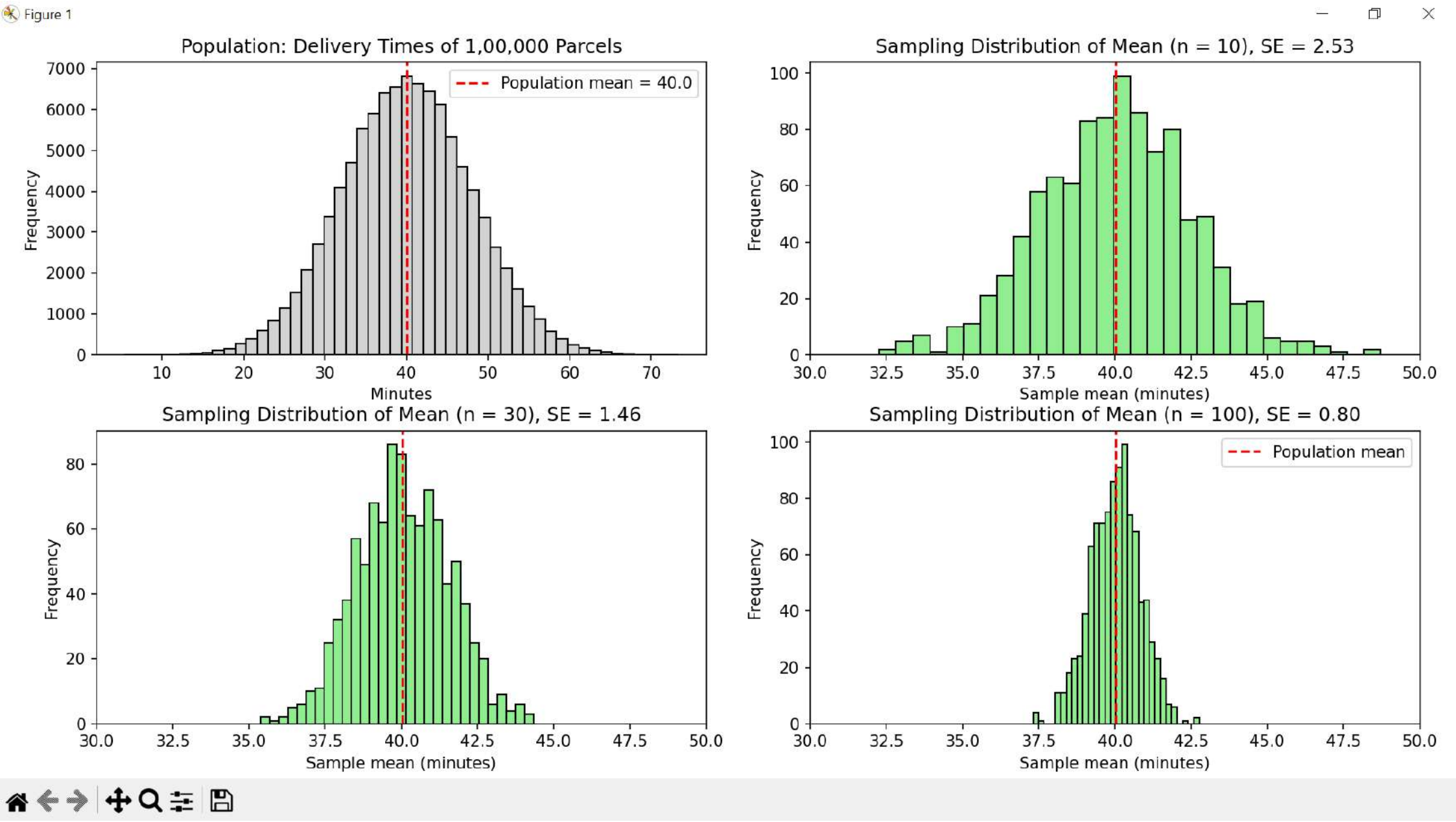
IDLE Shell 3.14.2

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Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/pings/Desktop/son.pu.py ==========
Population mean = 40.04 min, Population SD = 7.99 min

    n   Mean of sample means    SE = σ/√n   SD of sample means
   10                 40.03         2.53                 2.49
   30                 40.06         1.46                 1.52
  100                 40.01         0.80                 0.81
```

Ln: 5 Col: 0


Figure 1
Population: Delivery Times of 1,00,000 Parcels
Population mean = 40.0
Frequency
Minutes
Sampling Distribution of Mean (n = 10), SE = 2.53
Frequency
Sample mean (minutes)
Sampling Distribution of Mean (n = 30), SE = 1.46
Frequency
Sample mean (minutes)
Sampling Distribution of Mean (n = 100), SE = 0.80
Population mean
Frequency
Sample mean (minutes)